

LATIHAN P14 PEMROGRAMAN BERORIENTASI OBJEK

Nama : Torikh Abdullah Naser

NIM : 2410512031

Kelas : A-S1 Sistem Informasi

Koneksi.java

```
p14 > Koneksi.java > Koneksi > main(String[])
1 package p14;
2
3 import java.sql.Connection;
4 import java.sql.DriverManager;
5 import java.sql.SQLException;
6
7 public class Koneksi {
8     private static final String URL = "jdbc:mysql://localhost:3306/db_pbo";
9     private static final String USER = "root";
10    private static final String PASSWORD = "root";
11
12    public static Connection getConnection() throws SQLException {
13        return DriverManager.getConnection(URL, USER, PASSWORD);
14    }
15
16    Run | Debug
17    public static void main(String[] args) {
18        try (Connection conn = getConnection()) {
19            if (conn != null) {
20                System.out.println("Koneksi ke MySQL (Docker) BERHASIL!");
21            }
22        } catch (SQLException e) {
23            System.out.println("Koneksi gagal: " + e.getMessage());
24            e.printStackTrace();
25        }
26    }
27
28    > Generate Mermaid Diagram | Open Chat @mermaid-chart
```

```
... 10 more
PS D:\kuliah\smt 4\prak pbo> java -cp ".;p14/lib/*" p14.Koneksi
Koneksi ke MySQL (Docker) BERHASIL!
PS D:\kuliah\smt 4\prak pbo>
```

Mahasiswa.java

```
p14 > Mahasiswa.java > ...
1 package p14;
2
3 import java.sql.Connection;
4 import java.sql.Statement;
5
6 public class Mahasiswa {
7     Run | Debug
8     public static void main(String[] args) {
9         try (Connection conn = Koneksi.getConnection();
10             Statement stmt = conn.createStatement()) {
11
12             // Menggunakan Text Block (""" ) agar query SQL mudah dibaca
13             String sql = """
14                 CREATE TABLE IF NOT EXISTS mahasiswa (
15                     nim VARCHAR(10) PRIMARY KEY,
16                     nama VARCHAR(50),
17                     jurusan VARCHAR(50)
18                 )
19             """;
20
21             stmt.execute(sql);
22             System.out.println("Tabel mahasiswa berhasil dibuat di MySQL Docker!");
23
24         } catch (Exception e) {
25             System.out.println("Gagal membuat tabel: " + e.getMessage());
26             e.printStackTrace();
27         }
28     }
29
30     > Generate Mermaid Diagram | Open Chat @mermaid-chart
```

```
PS D:\kuliah\smt 4\prak pbo> java -cp ".;p14/lib/*" p14.Mahasiswa
Tabel mahasiswa berhasil dibuat di MySQL Docker!
PS D:\kuliah\smt 4\prak pbo>
```

InsertData.java

```
p14 > InsertData.java > {} p14
1  package p14;
2
3  import java.sql.Connection;
4  import java.sql.PreparedStatement;
5
6  public class InsertData {
7      Run | Debug
8      public static void main(String[] args) {
9
10         String sql = "INSERT INTO mahasiswa (nim, nama, jurusan) VALUES (?, ?, ?)"
11
12         // Data 10 mahasiswa
13         String[][] dataMahasiswa = {
14             {"M001", "Andi Pratama", "Informatika"},
15             {"M002", "Budi Santoso", "Sistem Informasi"},
16             {"M003", "Citra Lestari", "Teknik Komputer"},
17             {"M004", "Dewi Anggraini", "Informatika"},
18             {"M005", "Eko Wijaya", "Teknik Komputer"},
19             {"M006", "Fajar Nugroho", "Sistem Informasi"},
20             {"M007", "Gita Permata", "Informatika"},
21             {"M008", "Hendra Saputra", "Teknik Komputer"},
22             {"M009", "Intan Kusuma", "Sistem Informasi"},
23             {"M010", "Joko Prasetyo", "Informatika"}
24         };
25
26         try (Connection conn = Koneksi.getConnection();
27             PreparedStatement pstmt = conn.prepareStatement(sql)) {
28
29             // Loop insert 10 data menggunakan Batch
30             for (String[] mhs : dataMahasiswa) {
31                 pstmt.setString(1, mhs[0]); // nim
32                 pstmt.setString(2, mhs[1]); // nama
33                 pstmt.setString(3, mhs[2]); // jurusan
34                 pstmt.addBatch();
35             }
36
37             pstmt.executeBatch();
38             System.out.println("10 data mahasiswa berhasil dimasukkan!");
39
40         } catch (Exception e) {
41             e.printStackTrace();
42         }
43     }
44 }
```

```
● PS D:\kuliah\smt 4\prak pbo> java -cp ".;p14/lib/*" p14.InsertData
10 data mahasiswa berhasil dimasukkan!
○ PS D:\kuliah\smt 4\prak pbo>
```

LihatSemuaMahasiswa.java

```
p14 >  LihatSemuaMahasiswa.java > {} p14
1  package p14;
2
3  import java.sql.Connection;
4  import java.sql.ResultSet;
5  import java.sql.Statement;
6
7  public class LihatSemuaMahasiswa {
8      Run | Debug
9      public static void main(String[] args) {
10         String sql = "SELECT * FROM mahasiswa";
11         try (Connection conn = Koneksi.getConnection();
12             Statement stmt = conn.createStatement();
13             ResultSet rs = stmt.executeQuery(sql)) {
14
15             System.out.println("=== DATA SELURUH MAHASISWA ===");
16             while (rs.next()) {
17                 System.out.println(
18                     rs.getString("nim") + " - " +
19                     rs.getString("nama") + " - " +
20                     rs.getString("jurusan")
21                 );
22             }
23         } catch (Exception e) {
24             e.printStackTrace();
25         }
26     }
27 }
```

```
PS D:\kuliah\smt 4\prak pbo> java -cp ".;p14/lib/*" p14.LihatSemuaMahasiswa
=== DATA SELURUH MAHASISWA ===
M001 - Andi Pratama - Informatika
M002 - Budi Santoso - Sistem Informasi
M003 - Citra Lestari - Teknik Komputer
M004 - Dewi Anggraini - Informatika
M005 - Eko Wijaya - Teknik Komputer
M006 - Fajar Nugroho - Sistem Informasi
M007 - Gita Permata - Informatika
M008 - Hendra Saputra - Teknik Komputer
M009 - Intan Kusuma - Sistem Informasi
M010 - Joko Prasetyo - Informatika
PS D:\kuliah\smt 4\prak pbo> 
```

CariMahasiswaByJurusan.java

```
1 package p14;
2
3 import java.sql.Connection;
4 import java.sql.PreparedStatement;
5 import java.sql.ResultSet;
6
7 public class CariMahasiswaByJurusan { → Tab to Jump
8     Run | Debug
9     public static void main(String[] args) {
10         String sql = "SELECT * FROM mahasiswa WHERE jurusan = ?";
11         try (Connection conn = Koneksi.getConnection();
12             PreparedStatement pstmt = conn.prepareStatement(sql)) {
13             pstmt.setString(1, "Informatika");
14             ResultSet rs = pstmt.executeQuery();
15
16             System.out.println("=== MAHASISWA JURUSAN INFORMATIKA ===");
17             while (rs.next()) {
18                 System.out.println(
19                     rs.getString("nim") + " - " +
20                     rs.getString("nama") + " - " +
21                     rs.getString("jurusan")
22                 );
23             }
24         } catch (Exception e) {
25             e.printStackTrace();
26         }
27     }
28 }
```

```
PS D:\kuliah\smt 4\prak pbo> java -cp ".;p14/lib/*" p14.CariMahasiswaByJurusan
=== MAHASISWA JURUSAN INFORMATIKA ===
M001 - Andi Pratama - Informatika
M004 - Dewi Anggraini - Informatika
M007 - Gita Permata - Informatika
M010 - Joko Prasetyo - Informatika
PS D:\kuliah\smt 4\prak pbo>
```

UpdateMahasiswa.java

```
1 package p14;
2
3 import java.sql.Connection;
4 import java.sql.PreparedStatement;
5
6 public class UpdateMahasiswa {
7     Run | Debug
8     public static void main(String[] args) {
9         String sql = "UPDATE mahasiswa SET nama = ? WHERE nim = ?";
10        try (Connection conn = Koneksi.getConnection();
11            PreparedStatement pstmt = conn.prepareStatement(sql)) {
12            pstmt.setString(1, "Citra Larasati"); // Nama baru
13            pstmt.setString(2, "M003"); // NIM yang diubah
14
15            int affected = pstmt.executeUpdate();
16            System.out.println("Data diupdate: " + affected + " baris");
17
18        } catch (Exception e) {
19            e.printStackTrace();
20        }
21    }
22 }
```

```

PS D:\kuliah\smt 4\prak pbo> java -cp ".;p14/lib/*" p14.UpdateMahasiswa
Data diupdate: 1 baris
PS D:\kuliah\smt 4\prak pbo>

```

HitungMahasiswaPerJurusan

```

p14 > HitungMahasiswaPerJurusan.java > ...
1  package p14;
2
3  import java.sql.Connection;
4  import java.sql.PreparedStatement;
5  import java.sql.ResultSet;
6
7  public class HitungMahasiswaPerJurusan {
8      public static void main(String[] args) {
9          String sql = "SELECT jurusan, COUNT(*) AS total FROM mahasiswa GROUP BY jurusan";
10         try (Connection conn = Koneksi.getConnection();
11             PreparedStatement pstmt = conn.prepareStatement(sql)) {
12             ResultSet rs = pstmt.executeQuery();
13
14             System.out.println("=== TOTAL MAHASISWA PER JURUSAN ===");
15             while (rs.next()) {
16                 System.out.println(rs.getString("jurusan") + " = " + rs.getInt("total"));
17             }
18         } catch (Exception e) {
19             e.printStackTrace();
20         }
21     }
22 }
23
24

```

```

PS D:\kuliah\smt 4\prak pbo> java -cp ".;p14/lib/*" p14.HitungMahasiswaPerJurusan
=== TOTAL MAHASISWA PER JURUSAN ===
Informatika = 4
Sistem Informasi = 3
Teknik Komputer = 3
PS D:\kuliah\smt 4\prak pbo>

```